SKILL WEEK-2

STEP 1: Open Ubuntu

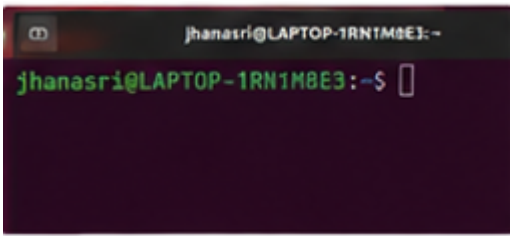
Open **Ubuntu** from the Start menu.

You should see:

jhanasri@LAPTOP-1RN1M8E3:~$

If you're somewhere else, go to your home folder:

cd ~



STEP 2: Create the OSSP folder

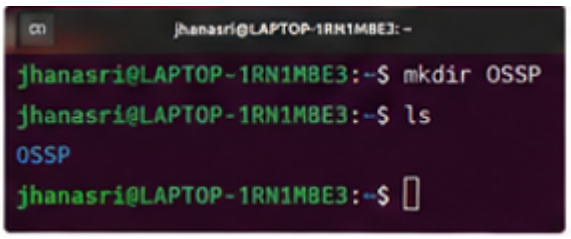
mkdir OSSP

Check:

ls

You should see

OSSP



STEP 3: Enter the folder

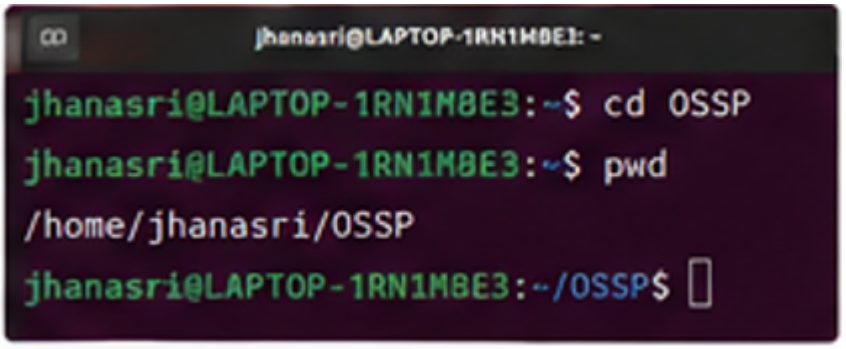
cd OSSP

Verify:

pwd

Output:

/home/jhanasri/OSSP



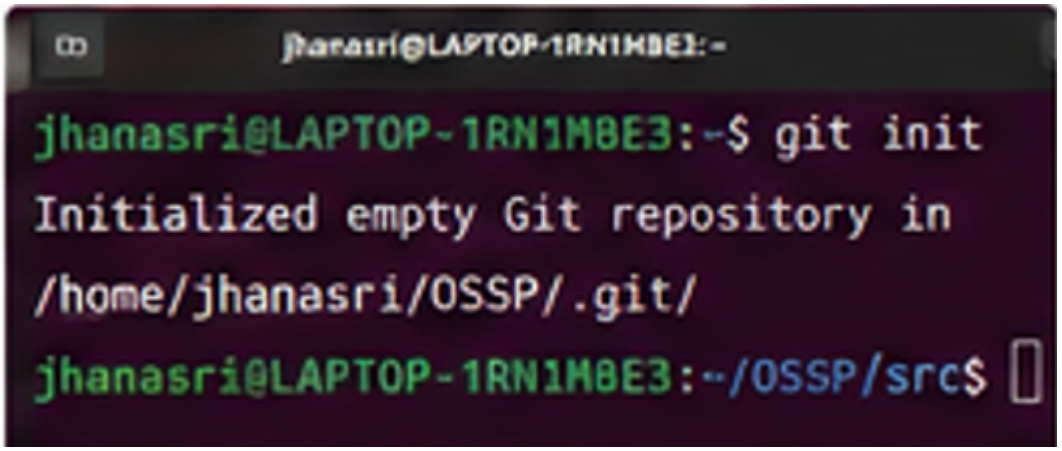
STEP 4: Initialize Git

git init

Expected output:

Initialized empty Git repository...

This creates the hidden `.git` directory automatically.



STEP 5: Create folders

Run one command:

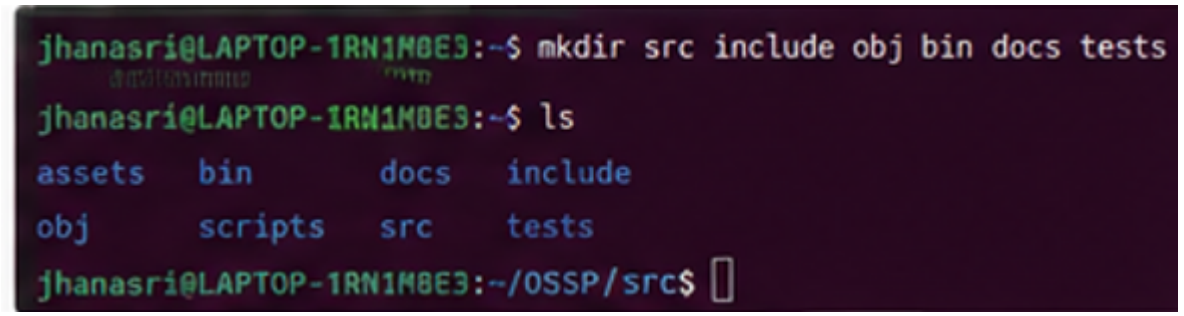
mkdir src include obj bin docs tests scripts assets

Check:

ls

You should see

assets
bin
docs
include
obj
scripts
src
Tests



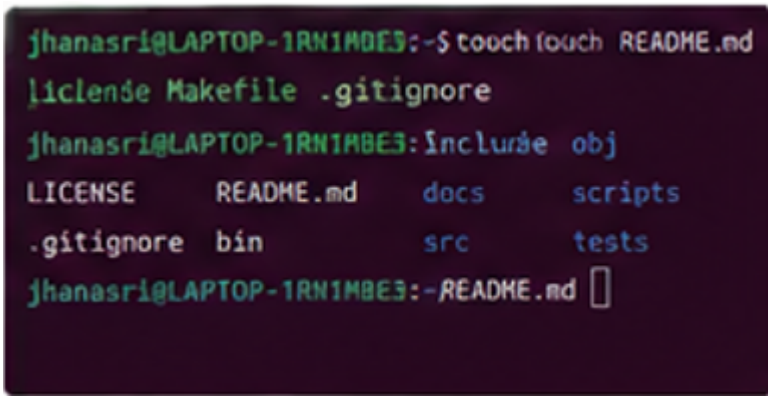
STEP 6: Create project files

Run:

touch README.md LICENSE Makefile .gitignore

Check:

Ls



STEP 7: Create source files

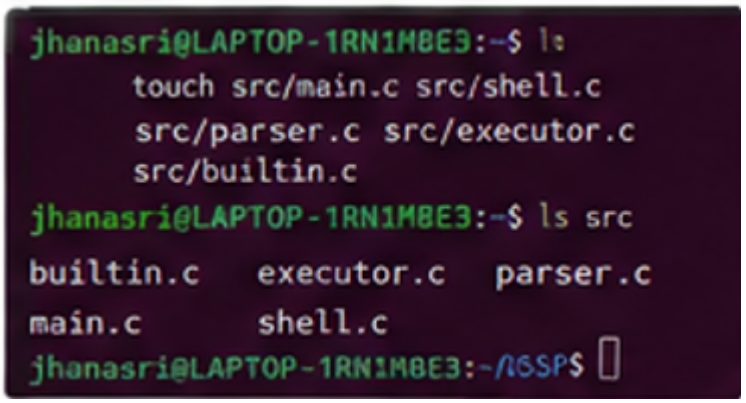
touch src/main.c
touch src/shell.c
touch src/parser.c
touch src/executor.c
touch src/builtin.c

Check:

ls src

Expected:

builtin.c
executor.c
main.c
parser.c
Shell.c



STEP 8: Create header files

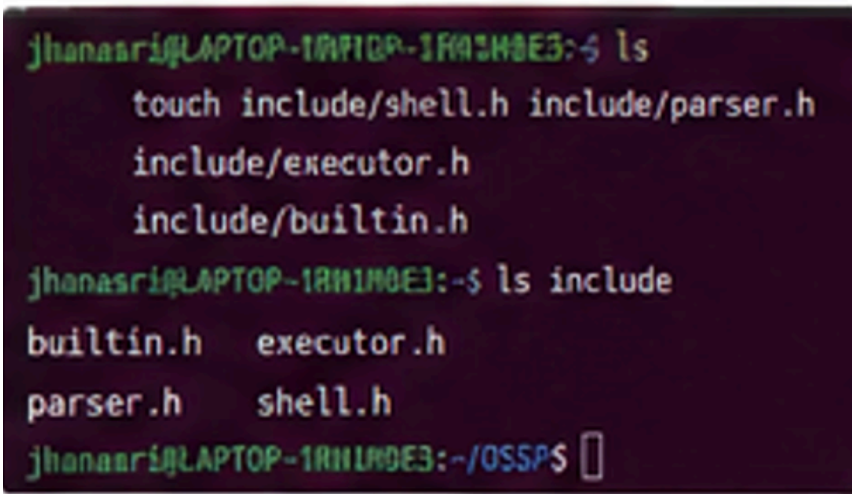
touch include/shell.h
touch include/parser.h
touch include/executor.h
touch include/builtin.h

Check:

ls include

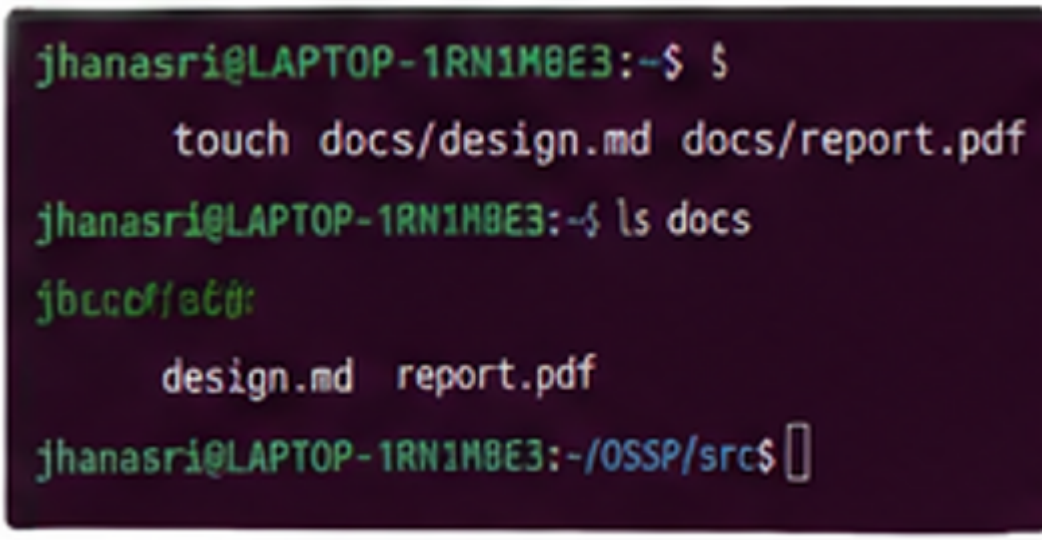
Expected:

builtin.h
executor.h
parser.h
Shell.h



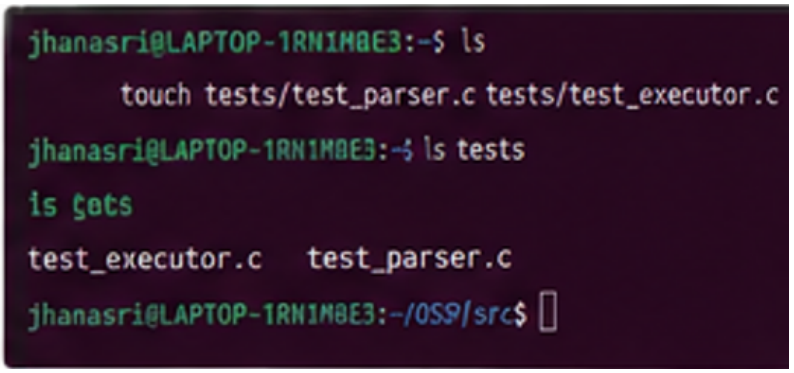
STEP 9: Create documentation files

touch docs/design.md
touch docs/report.pdf



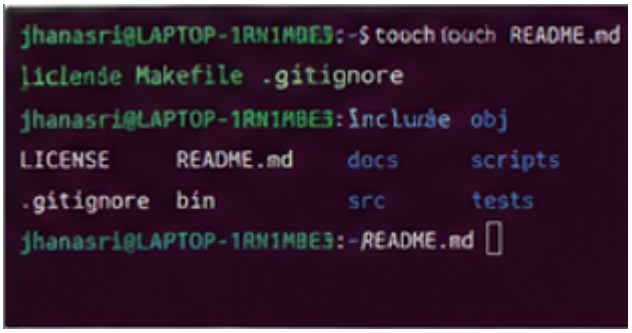
STEP 10: Create test files

touch tests/test_parser.c
touch tests/test_executor.c



STEP 11: Create helper script

touch scripts/run.sh



STEP 12: Create image placeholder

touch assets/architecture.png

The document lists this image under `assets/`; creating an empty placeholder file satisfies the required structure.

STEP 13: Check the complete structure

Run:

tree

If you get:

tree: command not found

Install it:

sudo apt install tree -y

Then run:

tree

You should see a structure similar to:

```
OSSP
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── tests
│   ├── test_executor.c
│   └── test_parser.c
├── .gitignore
├── LICENSE
├── Makefile
└── README.md
```

STEP 14: Verify Git

Run:

git status

You should see that Git has detected the new files.

Step 15: Open the `src` folder

cd ~/OSSP/src

Step 16: Create a new C file

touch fork_exec.c

Check:

ls

Step 17: Open the file

nano fork_exec.c

Step 18: Paste this code

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>


int main()

{

    char command[100];


    printf("Enter Linux Command: ");
```

```
scanf("%s", command);

pid_t pid = fork();

if(pid < 0)
{
    printf("Fork Failed!\n");
    return 1;
}
else if(pid == 0)
{
    printf("Child Process\n");
    printf("Child PID = %d\n", getpid());

    execlp(command, command, NULL);

    printf("Command Not Found!\n");
}
else
{
    wait(NULL);

    printf("Parent Process\n");
    printf("Parent PID = %d\n", getpid());
}
```

```
    return 0;
}
```

Step 19: Save the file

Press:

Ctrl + O

Press **Enter**

Then:

Ctrl + X

Step 20: Check whether GCC is installed

gcc --version

If you see a version number, continue.

If not, install it:

sudo apt install build-essential -y

Step 21: Compile the program

gcc fork_exec.c -o fork_exec

Step 22: Check the executable

ls

You should see:

fork_exec

Step 23: Run the program

./fork_exec

Step 24: Test with `ls`

Input:

ls

Step 25: Run again

./fork_exec

Input:

pwd

Step 26: Run again

./fork_exec

Input:

date

Step 27: Run again

./fork_exec

Input:

whoami

Step 28: View the process tree (optional but useful)

pstree

If `pstree` is not installed:

sudo apt install psmisc -y

Then run:

pstree

Step 29: Trace system calls (if required)

Run:

strace ./fork_exec

If `strace` is not installed:

sudo apt install strace -y

Then run the command again.

Step 30: Check Git status

cd ~/OSSP

git status